Deflection of a Standard PLA Filament Bracket

TM

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1 Introduction

A 3D-printed bracket is made to support the load of a surround-sound speaker weighing $P = 6.35\,lb$. Given that the elastic bending modulus, E, of Ender PLA+ filament is $331701.906\,psi$, with each member having a 20% filled triangular lattice ($\delta_{20\%} \approx 1.317988830848\delta_{100\%}$), estimate the deflection, δ, of point B as shown in Fig. (1):

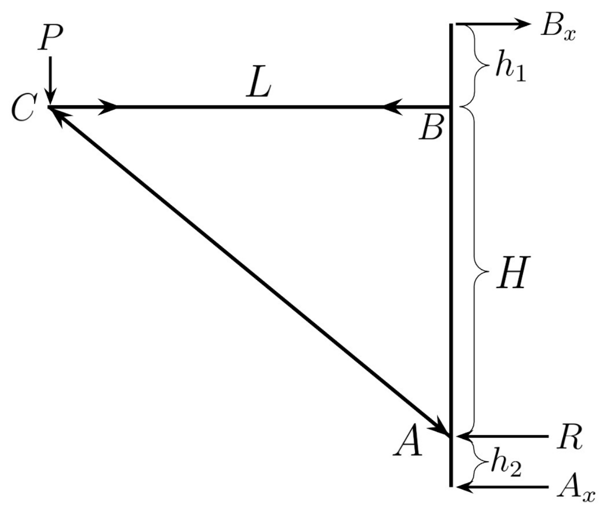

Figure 1: *A triangular truss supporting a load, P, by two drywall screws at points A_x and B_x.*

The following information is provided for Fig. (2):

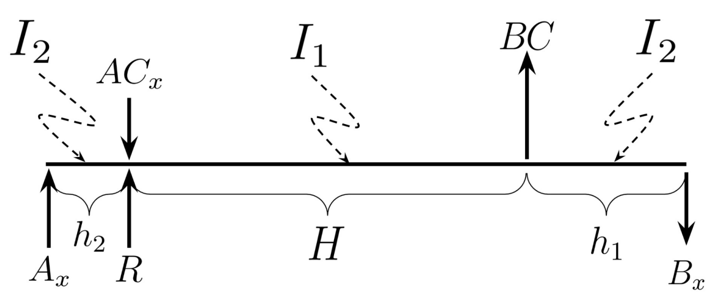

Figure 2: *Member AB from Fig. 1. Assume that point A is cantilevered and point B is pinned (can freely rotate).*

Givens:

- $P = 6.35\,lb$
- $E = 331701.906\,psi$

- $H = 4\,in.$
- $L = 5.5\,in.$
- $h_1 = 1.5\,in.$
- $h_2 = 1.0\,in.$
- $I_1 = 0.004\,in^4$ for $h_2 \leq x \leq H + h_2$
- $I_2 = 0.0005\,in^4$ for $0 \leq x \leq h_2$ and $H + h_2 \leq x \leq H + h_1 + h_2$

2 Solution

A quick statics analysis by the method of joints yields,

$$\begin{aligned} AC &= \frac{P\sqrt{L^2 + H^2}}{H} \\ AC_x &= \frac{PL}{H} \\ BC &= \frac{PL}{H} \\ A_x &= \frac{PL}{H + h_1 + h_2} \end{aligned}$$

the top screw's resistance, B_x, can be determined.

$$\circlearrowleft \sum_i^n M_A = 0 = PL - B_x \left(H + h_1 + h_2\right)$$

$$B_x = \frac{PL}{H + h_1 + h_2}$$

We invoke the method of superposition on member AB. Because the diagonal brace acts against the wall, the deflection between $x = 0$ and $x = h_2$ is negligible. By letting point A be a cantilever, we have a statically indeterminate system in which the reaction force, R, is unknown. Negating h_2, let $x = 0$ be the location of R.

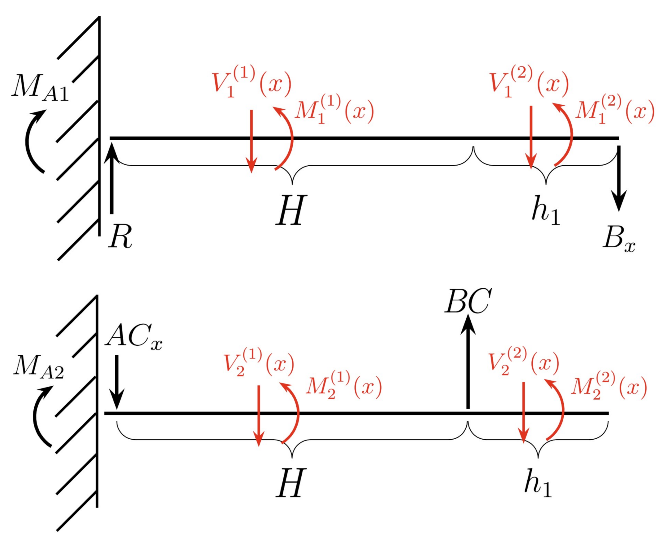

Figure 3: *Two separated problems each with their shear and moment functions denoted by the subscript. The superscript indicates the section pertaining to the inertias, I_1 and I_2.*

The deflection of the nth interval of x is,

$$\frac{\mathrm{d}^2\delta_i}{\mathrm{d}x^2} = \frac{\mathrm{d}\theta_i}{\mathrm{d}x} = \frac{S \cdot M^{(i)}(x)}{EI_i}$$

where E is the modulus of elasticity of the PLA plastic, I_n is the moment of inertia of the ith section, and S is a factor pertaining to the percent PLA filler in the structural member. Experimentally, for 20% filler of a triangular lattice, it was determined that $S \approx 1.339869$.

The deflection of the member AB given different inertia, I_n, in each interval x distance along it is an iterative process: determine the bending moment functions, solve the second-order deflection differential equations, and solve the coefficients given some assumed boundary conditions.

2.1 Case 1: Downward Deflection

We begin by determining the reacting moment at point A and force, B_x:

$$\uparrow^{+} \sum F_x = 0 = R - B_x$$
$$\therefore B_x = R$$

$$\circlearrowleft^{+} \sum M_A = 0 = -M_{A1} - B_x\,(H + h_1)$$
$$\therefore M_{A1} = -R\,(H + h_1)$$

For $0 \leq x \leq H$, the moment function along AB is determined to be

$$\circlearrowleft^{+} \sum M = 0 = M_1^{(1)}(x) - Rx - M_{A1}$$
$$\therefore M_1(x)^{(1)} = M_1^{(2)}(x) = Rx - R\,(H + h_1)$$

Given the boundary conditions, $\theta_1^{(1)}(0) = 0$, $\delta_1^{(1)}(0) = 0$,

$$\begin{aligned}
\theta_1^{(1)}(x) &= \frac{S}{EI_1} \int M_1(x)dx \\
&= \frac{S}{EI_1} \int (Rx - R\,(H + h_1))\,dx \\
&= \frac{S}{EI_1} \left[\frac{Rx^2}{2} - R\,(H + h_1)\,x + a_1\right] \quad , \quad \theta_1^{(1)}(0) = 0 \Rightarrow a_1 = 0 \\
\delta_1^{(1)}(x) &= \frac{S}{EI_1} \left[\frac{Rx^3}{6} - \frac{Rx^2}{2}\,(H + h_1) + a_2\right] \quad , \quad \delta_1^{(1)}(0) = 0 \Rightarrow a_2 = 0 \\
\therefore \delta_1^{(1)}(x) &= \frac{SR}{EI_1} \left[\frac{x^3}{6} - \frac{x^2}{2}\,(H + h_1)\right]
\end{aligned} \tag{1}$$

And,

$$\begin{aligned}
\theta_1^{(2)}(x) &= \frac{S}{EI_2} \int (Rx - R\,(H + h_1))\,dx \\
&= \frac{S}{EI_2} \left[\frac{Rx^2}{2} - R\,(H + h_1)\,x + b_1\right] \\
\theta_1^{(2)}(H) &= \theta_1^{(1)}(H) \Rightarrow b_1 = \left(-\frac{RH^2}{2} + RH(H + h_1)\right)\left(1 - \frac{I_2}{I_1}\right) \\
\delta_1^{(2)}(x) &= \frac{S}{EI_2} \left[\frac{Rx^3}{6} - \frac{R\,(H + h_1)\,x^2}{2} + \left(-\frac{RH^2}{2} + RH\,(H + h_1)\right)\left(1 - \frac{I_2}{I_1}\right) x + b_2\right] \\
\delta_1^{(1)}(H) &= \delta_1^{(2)}(H) \Rightarrow b_2 = \left(\frac{RH^3}{3} - \frac{RH^2\,(H + h_1)}{2}\right)\left(1 - \frac{I_2}{I_1}\right) \\
\therefore \delta_1^{(2)}(x) &= \frac{SR}{EI_2} \left[\frac{x^3}{6} - \frac{(H + h_1)\,x^2}{2} + \left(\frac{H^2}{2} + Hh_1\right)\left(1 - \frac{I_2}{I_1}\right) x - \left(\frac{H^3}{6} + \frac{H^2 h_1}{2}\right)\left(1 - \frac{I_2}{I_1}\right)\right]
\end{aligned} \tag{2}$$

2.2 Case 2: Upward Deflection

The reaction forces and moments on AB for case 2 is,

$$\circlearrowleft^{+} \sum M_A = 0 = -M_{A2} + BCH$$
$$\therefore M_{A2} = BCH$$

$$\uparrow^{+} \sum F_x = 0 = BC - AC_x$$
$$\therefore AC_x = BC$$

For $0 \leq x \leq H$

$$\circlearrowleft^{+} \sum M = 0 = M_2^{(1)}(x) + AC_x x - M_{A2}$$
$$\therefore M_2^{(1)}(x) = BC\,(H - x)$$

For $H \leq x \leq H + h_1$

$$\circlearrowleft^{+} \sum M = 0 = M_2^{(2)}(x) + AC_x x - M_{A2} - BC\,(x - H)$$
$$\therefore M_2^{(2)}(x) = 0$$

Given the following boundary conditions, we have,

$$\begin{aligned}
\theta_2^{(1)}(x) &= \frac{S}{EI_1} \int M_2^{(1)}(x) dx \\
&= \frac{S}{EI_1} \int BC\,(H - x)\, dx \\
&= \frac{S}{EI_1} \left[BC \left(-\frac{x^3}{6} + \frac{Hx^2}{2} \right) + c_1 \right] \quad , \quad \theta_2^{(1)}(0) = 0 \Rightarrow c_1 = 0 \\
\delta_2^{(1)}(x) &= \frac{S}{EI_1} \left[BC \left(-\frac{x^3}{6} + \frac{Hx^2}{2} \right) + c_2 \right] \quad , \quad \delta_2^{(1)}(0) = 0 \Rightarrow c_2 = 0 \\
\therefore \delta_2^{(1)}(x) &= \frac{SBC}{EI_1} \left(-\frac{x^3}{6} + \frac{Hx^2}{2} \right)
\end{aligned} \tag{3}$$

And,

$$\begin{aligned}
\theta_2^{(2)}(x) &= \frac{S}{EI_2} \int M_2^{(2)}(x) dx \\
&= \frac{S}{EI_2} \int 0 dx \\
&= \frac{S}{EI_2} d_1 \quad , \quad \theta_2^{(2)}(H) = \theta_2^{(1)}(H) \Rightarrow d_1 = BC \frac{H^2}{2} \frac{I_2}{I_1} \\
\delta_2^{(2)}(x) &= \frac{S}{EI_2} \left[BC \frac{H^2}{2} \frac{I_2}{I_1} x + d_2 \right] \quad , \quad \delta_2^{(2)}(H) = \delta_2^{(1)}(H) \Rightarrow d_2 = -BC \frac{H^3}{6} \frac{I_2}{I_1} \\
\therefore \delta_2^{(2)}(x) &= \frac{SBC}{EI_1} \left(\frac{H^2}{2} x - \frac{H^3}{6} \right)
\end{aligned} \tag{4}$$

2.3 Solving for the Reaction, R:

We seek a solution to R in EquationS (1) and (2) in the upward deflection such that,

$$\delta_1^2(H + h_1) + \delta_2^2(H + h_1) = 0$$

Working through the algebra, we obtain,

$$\frac{SBC}{EI_1}\left(\frac{H^2\left(H+h_1\right)}{2}-\frac{H^3}{6}\right)=-\frac{S}{EI_2}\left[\frac{R\left(H+h_1\right)^3}{6}-\frac{R\left(H+h_1\right)^3}{2}+\right.$$
$$\left.\left(\frac{RH^2}{2}+RHh_1\right)\left(1-\frac{I_2}{I_1}\right)\left(H+h_1\right)-\left(\frac{RH^3}{6}+\frac{RH^2h_1}{2}\right)\left(1-\frac{I_2}{I_1}\right)\right]$$
$$-BC\frac{I_2}{I_1}\left(\frac{H^2\left(H+h_1\right)}{2}-\frac{H^3}{6}\right)=R\left[-\frac{\left(H+h_1\right)^3}{3}+\left(\frac{H^2\left(H+h_1\right)}{2}+Hh_1\left(H+h_1\right)-\frac{H^3}{6}-\frac{H^2h_1}{2}\right)\left(1-\frac{I_2}{I_1}\right)\right]$$

and with $BC=\frac{PL}{H}$,

$$R=\frac{PLI_2}{I_1}\left(\frac{H\left(H+h_1\right)}{2}-\frac{H^2}{6}\right)\left[\frac{\left(H+h_1\right)^3}{3}+\left(1-\frac{I_2}{I_1}\right)\left(H^2\left(H+h_1\right)-H\left(H+h_1\right)^2-\frac{H^3}{3}\right)\right]^{-1}$$

3 Calculated vs. Experimental Results:

We may obtain a flexural modulus from Table 3 of (Caminero 2019) [1]. Because the bracket was printed with the triangle on its side, the "flat" orientation's flexural modulus is $E_f = 2.287\,GPa = 331,701.906\,psi$.

Four deflection tests were performed on beams with 20% infill, each having unique cross-sectional bases b, heights h, and lengths l. A $20lb$ dumbbell was placed on each beam's midpoint while supported at their ends, where deflection, $\delta_{20\%}$, was measured using a caliber.

The following unity ratios in Table (1) were obtained using Equation (5).

$$\delta_{100\%}=\frac{Pl^3}{48EI} \tag{5}$$

$b\,(in.)$	$h\,(in.)$	$l\,(in.)$	$I_x\,(in.^4)$	$\delta_{100\%}\,(in.)$	$\delta_{20\%}\,(in.)$	$\delta_{100\%}/\delta_{20\%}$
0.20	0.28	3.80	0.000365867	0.188	0.225	0.83731
0.46	0.28	4.65	0.000841493	0.150	0.240	0.62537
0.39	0.28	2.80	0.000713440	0.039	0.045	0.85891
0.97	0.46	5.88	0.000786799	0.032	0.046	0.71334

Table 1: Base b, height h, inertias I, calculated deflections $\delta_{100\%}$, measured deflections $\delta_{20\%}$, and unity ratios, $1/S_i$.

A bending increase factor may be found by,

$$\begin{aligned} S &= \frac{1}{n}\sum_{i=1}^{n}\left(\frac{\delta_{20\%}}{\delta_{100\%}}\right)_i \\ &= \frac{1}{4}\left[1.194+1.599+1.164+1.402\right] \\ &= 1.33986919414 \end{aligned} \tag{6}$$

Applying the 20 lb dumbbell directly on point B of member AB of the bracket, the measured deflection was approximately 0.13 in. Letting $BC = 20\,lb$ (or $P = 14.545\,lb$) and $S = 1.339\ldots$ in,

$$\delta^{(1)}(x)=\delta_1^{(1)}(x)+\delta_2^{(1)}(x) \tag{7}$$

$$\delta^{(2)}(x)=\delta_1^{(2)}(x)+\delta_2^{(2)}(x) \tag{8}$$

and finding the global maximum of either of these equations, the estimated deflection is $\delta_{100\%} = 0.0771883\,in$. This yields a unity ratio of 0.593756153846 due to the high deflection of the h_1 region and the lack of a cantilevered support at h_2.

Equations (7) and (8) predict that the 6.35 lb speaker will cause a deflection of $\delta_{20\%} = 0.033698\,in$.

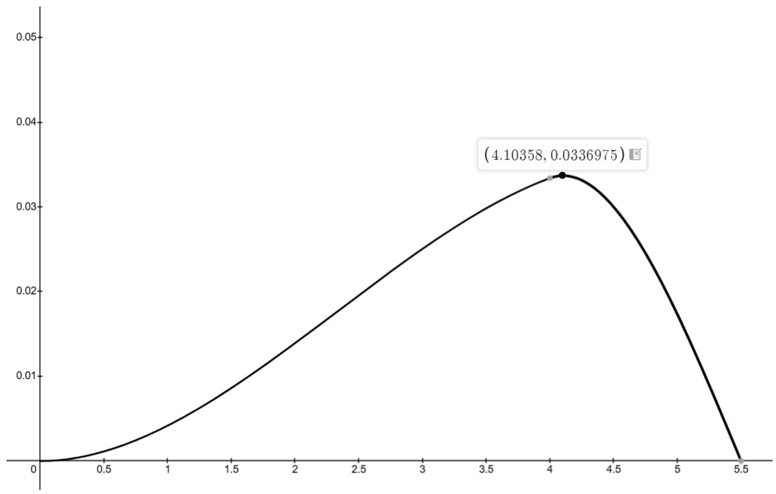

Figure 4: Predicted deflection due to the speaker along a vertical distance, x, along member AB.

References

[1] E. A. Miguel Caminero, *Additive manufacturing of pla-based composites using fused filament fabrication: Effect of graphene nanoplatelet reinforcement on mechanical properties, dimensional accuracy and texture*, MDPI, (2019).